

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

For a literature class, a student's essay needs to have no more than **5,000** words. If a student's essay already has **3,026** words, what is the maximum number of additional words the student can include in the essay?

Answer

- A. **1,974**
- B. **3,026**
- C. **5,000**
- D. **8,026**

Correct Answer: A

Rationale

Choice A is correct. It's given that a student's essay needs to have no more than **5,000** words. It's also given that the student's essay already has **3,026** words. Let x represent the number of additional words the student includes in the essay. After including x additional words, the essay will have a total of $3,026 + x$ words. Since the essay needs to have no more than **5,000** words, it follows that $3,026 + x \leq 5,000$. Subtracting **3,026** from both sides of this inequality yields $x \leq 1,974$. Thus, the maximum number of additional words the student can include in the essay is **1,974**.

Choice B is incorrect. This is the number of words already in the student's essay, rather than the maximum number of additional words the student can include.

Choice C is incorrect. This is the total number of words the essay can contain, rather than the maximum number of additional words the student can include.

Choice D is incorrect. This is the number of words the essay would contain if the student added **5,000** more words, rather than the maximum number of additional words the student can include.